

Chapter 10: Systems of Linear Differential Equations & Eigenvalues

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 10 (§10.1, §10.2)

1. Executive Overview & First-Principles Philosophy

In engineering systems, physical reality rarely isolates itself into a single degree of freedom or single state variable:

- **Coupled Mechanical Oscillators:** Multiple masses connected in series or parallel by springs and dampers (e.g., vehicle suspension models, multi-story buildings swaying in earthquakes) exhibit coupled displacements $x_1(t), x_2(t), \dots, x_n(t)$.
- **Multi-Loop Electrical Circuits:** Kirchhoff's voltage and current laws applied to multi-mesh networks containing inductors, capacitors, and resistors yield interconnected systems for loop currents $i_1(t), i_2(t), \dots, i_n(t)$.
- **Industrial Recirculation & Mixing Tanks:** Interconnected chemical reactors and brine tanks exchanging fluid via feed and recycle pipelines produce coupled solute mass rates $\frac{dx_1}{dt}, \frac{dx_2}{dt}$.

When these governing balance equations are formulated, they naturally assemble into a **System of First-Order Linear Differential Equations**.

Expressed in vector-matrix notation, an $n \times n$ system is written as:

$$\mathbf{X}'(t) = \mathbf{A}(t)\mathbf{X}(t) + \mathbf{F}(t)$$

where $\mathbf{X}(t)$ is the state vector, $\mathbf{A}(t)$ is the coefficient matrix, and $\mathbf{F}(t)$ is the external driving vector.

The profound mathematical beauty of linear systems theory is that **it converts calculus into linear algebra**:

- Finding the fundamental solutions of a constant-coefficient system $\mathbf{X}' = \mathbf{A}\mathbf{X}$ is completely equivalent to finding the **Eigenvalues λ and Eigenvectors $\mathbf{K}$** of the matrix $\mathbf{A}$.
- The eigenvalues govern the **temporal growth, decay, or oscillation frequency** of the system.
- The eigenvectors govern the **geometric mode shapes and directional invariant axes** in the phase plane.

2. Mathematical Framework & Linear Systems Theory (Zill §10.1)

2.1 Matrix Formulation of Linear Systems

A. Standard Normal Form

A system of n coupled first-order linear differential equations in n unknown functions $x_1(t), x_2(t), \dots, x_n(t)$ has the general form:

$$\frac{dx_1}{dt} = a_{11}(t)x_1 + a_{12}(t)x_2 + \dots + a_{1n}(t)x_n + f_1(t) \quad \frac{dx_2}{dt} = a_{21}(t)x_1 + a_{22}(t)x_2 + \dots + a_{2n}(t)x_n + f_2(t) \quad \vdots \quad \frac{dx_n}{dt} = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \dots + a_{nn}(t)x_n + f_n(t)$$

In compact matrix notation:

$$\mathbf{X}' = \mathbf{A}(t)\mathbf{X} + \mathbf{F}(t)$$

where:

$$\mathbf{X}(t) = \begin{pmatrix} x_1(t) & x_2(t) & \dots & x_n(t) \end{pmatrix}, \quad \mathbf{A}(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{pmatrix}, \quad \mathbf{F}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

- If $\mathbf{F}(t) \equiv \mathbf{0}$, the system is **homogeneous**: $\mathbf{X}' = \mathbf{A}(t)\mathbf{X}$.
- If $\mathbf{F}(t) \neq \mathbf{0}$, the system is **nonhomogeneous**.

B. Conversion of Higher-Order Scalar ODEs to First-Order Systems

Any single n -th order linear ODE can always be rewritten as an equivalent $n \times n$ first-order system.

- *Example*

: Consider $y'' + 3y' + 2y = 4 \cos(t)$. Define state variables: $x_1 = y, x_2 = y'$. Then:

$$x_1' = y' = x_2$$

$$x_2' = y'' = -2y - 3y' + 4 \cos(t) = -2x_1 - 3x_2 + 4 \cos(t)$$

In matrix form:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 & -2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \cos(t) \end{pmatrix}$$

C. Superposition Principle & Linear Independence

1. **Superposition Principle (Theorem 10.1.2):** If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k$ are solution vectors of the homogeneous system $\mathbf{X}' = \mathbf{A}\mathbf{X}$ on interval I , then any linear combination:

$$\mathbf{X}(t) = c_1\mathbf{X}_1(t) + c_2\mathbf{X}_2(t) + \dots + c_k\mathbf{X}_k(t)$$

is also a solution on I .

2. **Linear Independence & The Vector Wronskian:** A set of n solution vectors $\mathbf{X}_1, \dots, \mathbf{X}_n$ is linearly independent on I if and only if the **Wronskian determinant** is non-zero for every $t \in I$:

$$W(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n)(t) = \det [\mathbf{X}_1(t) \quad \mathbf{X}_2(t) \quad \dots \quad \mathbf{X}_n(t)] \neq 0$$

3. **Fundamental Matrix $\Phi(t)$:** The square matrix whose columns are n linearly independent solution vectors:

$$\Phi(t) = [\mathbf{X}_1(t) \quad \mathbf{X}_2(t) \quad \dots \quad \mathbf{X}_n(t)]$$

The general solution is then written compactly as:

$$\mathbf{X}(t) = \Phi(t)\mathbf{c}, \quad \text{where } \mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

2.2 Homogeneous Linear Systems with Constant Coefficients (Zill §10.2)

Consider the autonomous system where $\mathbf{A}$ is an $n \times n$ matrix of real constants:

$$\mathbf{X}' = \mathbf{A}\mathbf{X}$$

A. The Eigenvalue Trial Solution

By analogy with the scalar first-order equation $x' = ax \implies x(t) = ce^{at}$, we assume a trial vector solution:

$$\mathbf{X}(t) = \mathbf{K}e^{\lambda t}$$

where $\mathbf{K}$ is a non-zero constant vector $\begin{pmatrix} k_1 & k_2 \end{pmatrix}$ and λ is a scalar parameter. Differentiating with respect to t :

$$\mathbf{X}'(t) = \lambda \mathbf{K}e^{\lambda t}$$

Substituting into $\mathbf{X}' = \mathbf{A}\mathbf{X}$:

$$\lambda \mathbf{K}e^{\lambda t} = \mathbf{A}\mathbf{K}e^{\lambda t}$$

Dividing through by the non-zero scalar $e^{\lambda t}$:

$$\mathbf{A}\mathbf{K} = \lambda \mathbf{K} \iff (\mathbf{A} - \lambda \mathbf{I})\mathbf{K} = \mathbf{0}$$

This is the celebrated **Matrix Eigenvalue Problem**:

- λ is an **Eigenvalue** of the coefficient matrix $\mathbf{A}$.
- $\mathbf{K} \neq \mathbf{0}$ is the corresponding **Eigenvector**.

B. The Characteristic Equation

The homogeneous algebraic system $(\mathbf{A} - \lambda \mathbf{I})\mathbf{K} = \mathbf{0}$ possesses non-trivial solutions ($\mathbf{K} \neq \mathbf{0}$) if and only if the determinant of the coefficient matrix vanishes:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

For a 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

$$\lambda^2 - \text{Tr}(\mathbf{A})\lambda + \det(\mathbf{A}) = 0$$

where $\text{Tr}(\mathbf{A}) = a + d$ is the trace, and $\det(\mathbf{A}) = ad - bc$ is the determinant.

The roots λ_1, λ_2 of this quadratic equation govern three distinct physical and geometric behaviors.

2.3 The Three Foundational Cases & Phase Plane Topology

Case 1: Distinct Real Eigenvalues ($\lambda_1 \neq \lambda_2 \in \mathbb{R}$)

When the characteristic roots are real and distinct, each eigenvalue λ_i yields an independent real eigenvector $\mathbf{K}_i$ from $(\mathbf{A} - \lambda_i \mathbf{I})\mathbf{K}_i = \mathbf{0}$. The general solution is:

$$\mathbf{X}(t) = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \mathbf{K}_2 e^{\lambda_2 t}$$

Geometric Classifications in the Phase Plane:

- Asymptotically Stable Node (Sink)** ($\lambda_1 < 0$ and $\lambda_2 < 0$):
 - Both exponential terms decay to zero as $t \rightarrow \infty$. Every trajectory flows directly into the origin: $\lim_{t \rightarrow \infty} \mathbf{X}(t) = \mathbf{0}$.
 - Trajectories enter the origin tangent to the eigenvector corresponding to the **less negative** (slower decaying) eigenvalue.
- Unstable Node (Source)** ($\lambda_1 > 0$ and $\lambda_2 > 0$):
 - Both exponential terms grow without bound as $t \rightarrow \infty$. All trajectories flow away from the origin.
- Saddle Point (Always Unstable)** ($\lambda_1 > 0 > \lambda_2$ or opposite signs):
 - Trajectories starting along the stable eigenvector $\mathbf{K}_2$ (with $\lambda_2 < 0$) flow toward the origin as $t \rightarrow \infty$.
 - All other trajectories are deflected and asymptotically align with the unstable eigenvector $\mathbf{K}_1$ (with $\lambda_1 > 0$), fleeing to infinity.

corresponding entries in the vector of the left, we obtain the more familiar statement

$$x = c_1 e^{-t} + 3c_2 e^{4t}, y = -c_1 e^{-t} + 2c_2 e^{4t}.$$

As pointed out in Section 10.1, we can interpret these equations as parametric equations of a curve or **trajectory** in the xy -plane or **phase plane**. The three graphs shown in **FIGURE 10.2.1**, $x(t)$ in the tx -plane, $y(t)$ in the ty -plane, and the trajectory in the phase plane, correspond to the choice of constants $c_1 = c_2 = 1$ in the solution. A collection of trajectories in the phase plane as shown in **FIGURE 10.2.2** is said to be a **phase portrait** of the given linear system. What appears to be two red lines in Figure 10.2.2 are

Figure 10.2.2: Phase portrait of a 2x2 linear system with real eigenvalues. Trajectories flow along directional asymptotes defined by eigenvectors $\mathbf{K}_1$ and $\mathbf{K}_2$.

Case 2: Repeated Real Eigenvalues ($\lambda_1 = \lambda_2 = \lambda \in \mathbb{R}$)

When the characteristic equation has a repeated root of algebraic multiplicity 2:

- Subcase 2A: Complete / Non-Defective (Two Independent Eigenvectors):**
 - Occurs when $(\mathbf{A} - \lambda \mathbf{I}) = \mathbf{0}$, meaning $\mathbf{A} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda \end{pmatrix}$.
 - Any two non-collinear vectors serve as eigenvectors.
 - General solution: $\mathbf{X}(t) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{\lambda t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{\lambda t}$.
 - The phase portrait is a **Proper Node (Star Point)** where trajectories are straight radial rays.

- Subcase 2B: Defective (Only One Linearly Independent Eigenvector $\mathbf{K}$):**

- The first solution vector is:

$$\mathbf{X}_1(t) = \mathbf{K} e^{\lambda t}$$

- The second linearly independent solution **requires a generalized eigenvector $\mathbf{P}$** :

$$\mathbf{X}_2(t) = \mathbf{K} t e^{\lambda t} + \mathbf{P} e^{\lambda t}$$

- Derivation of the Generalized Eigenvector Equation:** Differentiate $\mathbf{X}_2(t)$:

$$\mathbf{X}_2'(t) = \mathbf{K} e^{\lambda t} + \lambda \mathbf{K} t e^{\lambda t} + \lambda \mathbf{P} e^{\lambda t}$$

Substitute $\mathbf{X}_2$ into $\mathbf{X}' = \mathbf{A}\mathbf{X}$:

$$\mathbf{K} e^{\lambda t} + \lambda \mathbf{K} t e^{\lambda t} + \lambda \mathbf{P} e^{\lambda t} = \mathbf{A} \mathbf{K} t e^{\lambda t} + \mathbf{A} \mathbf{P} e^{\lambda t}$$

Equating coefficients of $t e^{\lambda t}$ and $e^{\lambda t}$:

$$\mathbf{A} \mathbf{K} = \lambda \mathbf{K} \implies (\mathbf{A} - \lambda \mathbf{I}) \mathbf{K} = \mathbf{0} \quad (\text{Eigenvector equation})$$

$$\mathbf{A} \mathbf{P} - \lambda \mathbf{P} = \mathbf{K} \implies (\mathbf{A} - \lambda \mathbf{I}) \mathbf{P} = \mathbf{K} \quad (\text{Generalized Eigenvector equation})$$

- The general solution is:

$$\mathbf{X}(t) = c_1 \mathbf{K} e^{\lambda t} + c_2 (\mathbf{K}t + \mathbf{P}) e^{\lambda t}$$

- The phase portrait is an **Improper (Degenerate) Node**.

Case 3: Complex Conjugate Eigenvalues ($\lambda = \alpha \pm i\beta, \beta \neq 0$)

Because $\mathbf{A}$ is real, complex eigenvalues always occur in conjugate pairs:

$$\lambda_1 = \alpha + i\beta, \quad \lambda_2 = \alpha - i\beta$$

The eigenvector $\mathbf{K}_1$ associated with λ_1 will have complex entries. Decompose $\mathbf{K}_1$ into real and imaginary vector components:

$$\mathbf{K}_1 = \mathbf{B}_1 + i\mathbf{B}_2, \quad \text{where } \mathbf{B}_1 = \text{Re}(\mathbf{K}_1), \quad \mathbf{B}_2 = \text{Im}(\mathbf{K}_1)$$

The formal complex solution is:

$$\mathbf{W}(t) = \mathbf{K}_1 e^{(\alpha + i\beta)t} = (\mathbf{B}_1 + i\mathbf{B}_2) e^{\alpha t} (\cos(\beta t) + i \sin(\beta t))$$

Expanding the product:

$$\mathbf{W}(t) = e^{\alpha t} [(\mathbf{B}_1 \cos(\beta t) - \mathbf{B}_2 \sin(\beta t)) + i(\mathbf{B}_2 \cos(\beta t) + \mathbf{B}_1 \sin(\beta t))]$$

By the Superposition Principle, the **real part** $\text{Re}(\mathbf{W})$ and **imaginary part** $\text{Im}(\mathbf{W})$ form two real linearly independent solutions:

$$\mathbf{X}_1(t) = e^{\alpha t} [\mathbf{B}_1 \cos(\beta t) - \mathbf{B}_2 \sin(\beta t)]$$

$$\mathbf{X}_2(t) = e^{\alpha t} [\mathbf{B}_2 \cos(\beta t) + \mathbf{B}_1 \sin(\beta t)]$$

The general real solution is:

$$\mathbf{X}(t) = c_1 \mathbf{X}_1(t) + c_2 \mathbf{X}_2(t)$$

Geometric Classifications in the Phase Plane:

- Center (Neutral Stability)** ($\alpha = 0$, pure imaginary $\lambda = \pm i\beta$):
 - Trajectories form closed concentric ellipses around the origin. The system oscillates indefinitely with natural frequency $\omega = \beta$ without decaying or growing.
- Stable Spiral Point (Spiral Sink)** ($\alpha < 0$):
 - The factor $e^{\alpha t} \rightarrow 0$ dampens the oscillations. Trajectories spiral inward toward the origin as $t \rightarrow \infty$.
- Unstable Spiral Point (Spiral Source)** ($\alpha > 0$):
 - The factor $e^{\alpha t} \rightarrow \infty$ amplifies the oscillations. Trajectories spiral outward away from the origin.

Determining Direction of Rotation (Clockwise vs. Counterclockwise)

: Test the velocity vector $\mathbf{X}' = \mathbf{A}\mathbf{X}$ at the test point $(1, 0)^T$ on the positive x_1 -axis:

$$\mathbf{X}' = \begin{pmatrix} a_{11} & a_{12} & a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$$

- If $a_{21} > 0$: $x_2' > 0 \implies$ velocity points upward $\implies$ **Counterclockwise** rotation.
- If $a_{21} < 0$: $x_2' < 0 \implies$ velocity points downward $\implies$ **Clockwise** rotation.

3. Master Classification Matrix for 2×2 Linear Systems

Let $\tau = \text{Tr}(\mathbf{A}) = a + d$ and $\Delta = \det(\mathbf{A}) = ad - bc$. The characteristic equation is $\lambda^2 - \tau\lambda + \Delta = 0$. The discriminant is:

$$D = \tau^2 - 4\Delta$$

Condition on Δ, τ, D	Eigenvalues λ_1, λ_2	Stability Type	Phase Portrait Topology
$\Delta < 0$	Real, opposite signs ($\lambda_1 > 0 > \lambda_2$)	Unstable	Saddle Point
$\Delta > 0, D > 0, \tau < 0$	Real, distinct, both negative ($\lambda_2 < \lambda_1 < 0$)	Asymptotically Stable	Stable Node (Sink)
$\Delta > 0, D > 0, \tau > 0$	Real, distinct, both positive ($\lambda_1 > \lambda_2 > 0$)	Unstable	Unstable Node (Source)
$\Delta > 0, D = 0, \tau < 0$	Real, repeated, negative ($\lambda = \tau/2 < 0$)	Asymptotically Stable	Degenerate / Star Node
$\Delta > 0, D = 0, \tau > 0$	Real, repeated, positive ($\lambda = \tau/2 > 0$)	Unstable	Degenerate / Star Node
$\Delta > 0, D < 0, \tau < 0$	Complex conjugate with $\alpha < 0$ ($\lambda = \alpha \pm i\beta$)	Asymptotically Stable	Stable Spiral (Sink)
$\Delta > 0, D < 0, \tau > 0$	Complex conjugate with $\alpha > 0$ ($\lambda = \alpha \pm i\beta$)	Unstable	Unstable Spiral (Source)
$\Delta > 0, \tau = 0$ ($D < 0$)	Pure imaginary ($\lambda = \pm i\sqrt{\Delta}$)	Neutrally Stable	Center (Ellipses)

4. Comprehensive Step-by-Step Problem Walkthroughs

4.1 Problem 1: Real Distinct Eigenvalues IVP (Saddle Point)

Problem Statement: Solve the initial-value problem:

$$\mathbf{X}' = \begin{pmatrix} 1 & 3 & 5 \\ 3 & 3 & 3 \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

Classify the critical point at the origin and describe its phase portrait.

Step 1: Find the Eigenvalues

Compute $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$:

$$\begin{aligned} \det \begin{pmatrix} 1 - \lambda & 3 & 5 \\ 3 & 3 - \lambda \end{pmatrix} &= (1 - \lambda)(3 - \lambda) - (3)(5) = 0 \\ \lambda^2 - 4\lambda + 3 - 15 &= \lambda^2 - 4\lambda - 12 = 0 \end{aligned}$$

Factor the quadratic equation:

$$(\lambda - 6)(\lambda + 2) = 0 \implies \lambda_1 = 6, \quad \lambda_2 = -2$$

Since $\lambda_1 > 0$ and $\lambda_2 < 0$, the origin $(0, 0)$ is an **Unstable Saddle Point**.

Step 2: Find the Eigenvector $\mathbf{K}_1$ for $\lambda_1 = 6$

Substitute $\lambda_1 = 6$ into $(\mathbf{A} - 6\mathbf{I})\mathbf{K}_1 = \mathbf{0}$:

$$\begin{pmatrix} 1 - 6 & 3 & 5 \\ 3 & 3 - 6 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} -5 & 3 & 5 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first row:

$$-5k_1 + 3k_2 = 0 \implies 3k_2 = 5k_1 \implies k_2 = \frac{5}{3}k_1$$

Choosing $k_1 = 3$ gives $k_2 = 5$:

$$\mathbf{K}_1 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

The first fundamental solution is:

$$\mathbf{X}_1(t) = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

Step 3: Find the Eigenvector $\mathbf{K}_2$ for $\lambda_2 = -2$

Substitute $\lambda_2 = -2$ into $(\mathbf{A} - (-2)\mathbf{I})\mathbf{K}_2 = (\mathbf{A} + 2\mathbf{I})\mathbf{K}_2 = \mathbf{0}$:

$$\begin{pmatrix} 1 - (-2) & 3 & 5 \\ 3 & 3 - (-2) \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 5 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first row:

$$3k_1 + 3k_2 = 0 \implies k_1 + k_2 = 0 \implies k_2 = -k_1$$

Choosing $k_1 = 1$ gives $k_2 = -1$:

$$\mathbf{K}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The second fundamental solution is:

$$\mathbf{X}_2(t) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$$

Step 4: Assemble the General Solution

$$\mathbf{X}(t) = c_1 \mathbf{X}_1(t) + c_2 \mathbf{X}_2(t) = c_1 \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$$

Step 5: Enforce the Initial Condition $\mathbf{X}(0) = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$

At $t = 0$:

$$\mathbf{X}(0) = c_1 \begin{pmatrix} 3 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

This produces the 2×2 algebraic system:

$$3c_1 + c_2 = 2 \quad \text{--- (Eq. 1)} \quad 5c_1 - c_2 = 6 \quad \text{--- (Eq. 2)}$$

Add (Eq. 1) and (Eq. 2) directly:

$$(3c_1 + 5c_1) + (c_2 - c_2) = 2 + 6 \implies 8c_1 = 8 \implies c_1 = 1$$

Substitute $c_1 = 1$ into (Eq. 1):

$$3(1) + c_2 = 2 \implies c_2 = 2 - 3 = -1$$

Step 6: Final Solution

$$\mathbf{X}(t) = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t} - \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$$

In component form:

$$x_1(t) = 3e^{6t} - e^{-2t}, \quad x_2(t) = 5e^{6t} + e^{-2t}$$

4.2 Problem 2: Defective Repeated Eigenvalues (Generalized Eigenvector)

Problem Statement: Find the general solution of the linear system:

$$\mathbf{X}' = \begin{pmatrix} 3 & -18 & 2 & -9 \end{pmatrix} \mathbf{X}$$

Step 1: Find the Eigenvalues

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 3 - \lambda & -18 & 2 & -9 - \lambda \end{pmatrix} = (3 - \lambda)(-9 - \lambda) - (-18)(2) = 0$$

$$\lambda^2 + 6\lambda - 27 + 36 = \lambda^2 + 6\lambda + 9 = 0$$

$$(\lambda + 3)^2 = 0 \implies \lambda = -3 \quad (\text{Multiplicity } 2)$$

Step 2: Find the First Eigenvector $\mathbf{K}$

Substitute $\lambda = -3$ into $(\mathbf{A} + 3\mathbf{I})\mathbf{K} = \mathbf{0}$:

$$\begin{pmatrix} 3 - (-3) & -18 & 2 & -9 - (-3) \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 6 & -18 & 2 & -6 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Both rows reduce to:

$$2k_1 - 6k_2 = 0 \implies k_1 = 3k_2$$

Choosing $k_2 = 1$ yields $k_1 = 3$:

$$\mathbf{K} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Since the eigenspace has dimension 1 (only one independent eigenvector), the matrix is **defective**. The first solution is:

$$\mathbf{X}_1(t) = \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{-3t}$$

Step 3: Find the Generalized Eigenvector $\mathbf{P}$

Solve $(\mathbf{A} - \lambda \mathbf{I})\mathbf{P} = \mathbf{K}$, which is $(\mathbf{A} + 3\mathbf{I})\mathbf{P} = \mathbf{K}$:

$$\begin{pmatrix} 6 & -18 & 2 & -6 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Both equations state:

$$2p_1 - 6p_2 = 1 \implies 2p_1 = 1 + 6p_2 \implies p_1 = \frac{1}{2} + 3p_2$$

We are free to choose any convenient real value for p_2 . Setting $p_2 = 0$ gives $p_1 = \frac{1}{2}$:

$$\mathbf{P} = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$$

Step 4: Construct the Second Solution $\mathbf{X}_2(t)$

$$\mathbf{X}_2(t) = \mathbf{K}te^{-3t} + \mathbf{P}e^{-3t} = \left[\begin{pmatrix} 3 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} \right] e^{-3t}$$

Step 5: Assemble the General Solution

$$\mathbf{X}(t) = c_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{-3t} + c_2 \left[\begin{pmatrix} 3 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} \right] e^{-3t}$$

In component form:

$$x_1(t) = c_1(3e^{-3t}) + c_2 \left(3t + \frac{1}{2} \right) e^{-3t}$$

$$x_2(t) = c_1 e^{-3t} + c_2 t e^{-3t}$$

Stability

: Since $\lambda = -3 < 0$, the origin is an **Asymptotically Stable Degenerate Node**.

4.3 Problem 3: Complex Conjugate Eigenvalues (Center / Spiral)

Problem Statement: Find the general solution of the system:

$$\mathbf{X}' = \begin{pmatrix} 2 & 8 & -1 & -2 \end{pmatrix} \mathbf{X}$$

Step 1: Find the Eigenvalues

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & 8 & -1 & -2 - \lambda \end{pmatrix} = (2 - \lambda)(-2 - \lambda) - (8)(-1) = 0$$

$$\lambda^2 - 4 + 8 = \lambda^2 + 4 = 0 \implies \lambda = \pm 2i$$

Here $\alpha = 0$ and $\beta = 2$. Because $\alpha = 0$, the critical point is a **Neutrally Stable Center** (closed periodic trajectories).

Step 2: Find the Complex Eigenvector for $\lambda_1 = 2i$

Substitute $\lambda_1 = 2i$ into $(\mathbf{A} - 2i\mathbf{I})\mathbf{K} = \mathbf{0}$:

$$\begin{pmatrix} 2 - 2i & 8 & -1 & -2 - 2i \end{pmatrix} \begin{pmatrix} k_1 & k_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

From the second row:

$$-k_1 - (2 + 2i)k_2 = 0 \implies k_1 = -(2 + 2i)k_2$$

Choosing $k_2 = -1$ gives $k_1 = 2 + 2i$:

$$\mathbf{K}_1 = \begin{pmatrix} 2 + 2i & -1 \end{pmatrix}$$

Split $\mathbf{K}_1$ into real and imaginary vector parts:

$$\mathbf{K}_1 = \begin{pmatrix} 2 & -1 \end{pmatrix} + i \begin{pmatrix} 2 & 0 \end{pmatrix} \implies \mathbf{B}_1 = \begin{pmatrix} 2 & -1 \end{pmatrix}, \quad \mathbf{B}_2 = \begin{pmatrix} 2 & 0 \end{pmatrix}$$

Step 3: Construct the Two Real Linearly Independent Solutions

Using $\alpha = 0, \beta = 2$:

$$\mathbf{X}_1(t) = \mathbf{B}_1 \cos(2t) - \mathbf{B}_2 \sin(2t) = \begin{pmatrix} 2 & -1 \end{pmatrix} \cos(2t) - \begin{pmatrix} 2 & 0 \end{pmatrix} \sin(2t) = \begin{pmatrix} 2 \cos(2t) - 2 \sin(2t) & -\cos(2t) \end{pmatrix} \quad \mathbf{X}_2(t) = \mathbf{B}_2 \cos(2t) + \mathbf{B}_1 \sin(2t) = \begin{pmatrix} 2 & 0 \end{pmatrix} \cos(2t) + \begin{pmatrix} 2 & -1 \end{pmatrix} \sin(2t)$$

Step 4: Assemble the General Solution

$$\mathbf{X}(t) = c_1 \begin{pmatrix} 2 \cos(2t) - 2 \sin(2t) & -\cos(2t) \end{pmatrix} + c_2 \begin{pmatrix} 2 \cos(2t) + 2 \sin(2t) & \sin(2t) \end{pmatrix}$$

Direction of Rotation

: At test point $(1, 0)^T$, $\mathbf{X}' = \mathbf{A} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \end{pmatrix}$. Since the vertical velocity is $x'_2 = -1 < 0$, the flow on the positive x_1 -axis points downward $\implies$ **Clockwise Elliptical Orbits**.

5. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Dropping the Generalized Vector $\mathbf{P}$ in Repeated Roots:** When solving defective systems with repeated eigenvalues, writing $\mathbf{X}_2(t) = \mathbf{K}te^{\lambda t}$ is **completely wrong!** Substituting that into $\mathbf{X}' = \mathbf{A}\mathbf{X}$ yields $\mathbf{K}e^{\lambda t} = \mathbf{0}$, which is a contradiction. You **must** solve $(\mathbf{A} - \lambda \mathbf{I})\mathbf{P} = \mathbf{K}$ and include $\mathbf{P}$: $\mathbf{X}_2 = (\mathbf{K}t + \mathbf{P})e^{\lambda t}$.
- **⚠ Trap 2: Complex Eigenvector Imaginary Sign Flips:** Remember that $\mathbf{X}_1 = \mathbf{B}_1 \cos(\beta t) - \mathbf{B}_2 \sin(\beta t)$. Notice the **minus sign** in front of $\mathbf{B}_2 \sin(\beta t)$! It arises because $i \cdot i = -1$ when expanding the product $(\mathbf{B}_1 + i\mathbf{B}_2)(\cos \beta t + i \sin \beta t)$.
- **⚠ Trap 3: Sign Errors in Matrix Characteristic Determinants:** For $\mathbf{A} = \begin{pmatrix} a & b & c & d \end{pmatrix}$, $\det(\mathbf{A} - \lambda \mathbf{I}) = \lambda^2 - (a + d)\lambda + (ad - bc)$. Students often write $\lambda^2 + (a + d)\lambda$ or mess up the signs of $ad - bc$ when b or c is negative. Always double-check by calculating $\det$ by hand!
- **⚠ Trap 4: Rotation Direction in Spiral Phase Portraits:** Never guess clockwise vs counterclockwise. Always evaluate $\mathbf{A} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} \end{pmatrix}$. The sign of a_{21} directly gives the vertical velocity x'_2 : negative means downward (clockwise), positive means upward (counterclockwise).